# Stage 6. Complete the upgrade

AFF and FAS Controller Upgrade

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# Stage 6. Complete the upgrade

## Stage 6. Complete the upgrade

During stage 6, ensure that the new nodes are set up correctly. If one of the new nodes has a unified target adapter, you must restore any port configurations and might need to change the personality of the adapter. You should also set up Storage Encryption if the new nodes are encryption-enabled. You should also decommission the old nodes.

**Steps**

1. Ensure that the new controllers are set up correctly
2. Set up Storage Encryption on the new controller module
3. Set up NetApp Encryption on the new controller module
4. Decommission the old system
5. Resume SnapMirror operations

**For MetroCluster FC configuration**

For MetroCluster FC configuration, you must replace the disaster recovery/failover site nodes as soon as possible. Mismatch in controller models within a MetroCluster is not supported because controller model mismatch can cause disaster recovery mirroring to go offline. Use the command `-skip-metrocluster-check true` option to bypass MetroCluster checks when you are replacing nodes at second site.

## Confirm that the new controllers are set up correctly

To confirm correct setup, you must enable the HA pair. You must also verify that node3 and node4 can access each other's storage and that neither owns data LIFs belonging to other nodes on the cluster. In addition, you must confirm that node3 owns node1's aggregates and that node4 owns node2's aggregates, and that the volumes for both nodes are online.

**Steps**

1. After the post-checks of node2, the storage failover and cluster HA pair for the node2 cluster are enabled. When the operation is done, both nodes show as completed and the system performs some cleanup operations.
2. Verify that storage failover is enabled:

   ```
   storage failover show
   ```

   The following example shows the output of the command when storage failover is enabled:

```
cluster::> storage failover show
                   Takeover
Node      Partner  Possible   State Description
-------- -----    -------    ------------------
node3    node4    true       Connected to node4
node4    node3    true       Connected to node3
```

3. Verify that node3 and node4 belong to the same cluster by using the following command and examining the output:

   `cluster show`

4. Verify that node3 and node4 can access each other's storage by using the following command and examining the output:

   `storage failover show -fields local-missing-disks, partner-missing-disks`

5. Verify that neither node3 nor node4 owns data LIFs home-owned by other nodes in the cluster by using the following command and examining the output:

   `network interface show`

   If neither node3 or node4 owns data LIFs home-owned by other nodes in the cluster, revert the data LIFs to their home owner:

   `network interface revert`

6. Verify that node3 owns the aggregates from node1 and that node4 owns the aggregates from node2:

   `storage aggregate show -owner-name node3`

   `storage aggregate show -owner-name node4`

7. Determine whether any volumes are offline:

   `volume show -node node3 -state offline`

   `volume show -node node4 -state offline`

8. If any volumes are offline, compare them with the list of offline volumes that you captured in the section Prepare the nodes for upgrade, and bring online any of the offline volumes, as required, by using the following command, once for each volume:

   `volume online -vserver vserver_name -volume volume_name`

9. Install new licenses for the new nodes by using the following command for each node:

   `system license add -license-code license_code,license_code,license_code…`

   The license-code parameter accepts a list of 28 upper-case alphabetic character keys. You can add one license at a time, or you can add multiple licenses at once, separating each license key by a comma.

10. Remove all of the old licenses from original nodes by using one of the following commands:

    `system license clean-up -unused -expired`

    `system license delete -serial-number node_serial_number -package licensable_package`

    - Delete all expired licenses:

      `system license clean-up -expired`

    - Delete all unused licenses:

      `system license clean-up -unused`

    - Delete a specific license from a cluster by using the following commands on the nodes:

      `system license delete -serial-number node1_serial_number -package *`

      `system license delete -serial-number node2_serial_number -package *`

    The following output is displayed:

    ```
    Warning: The following licenses will be removed:
    <list of each installed package>
    Do you want to continue? {y|n}: y
    ```

    Enter `y` to remove all of the packages.

11. Verify that the licenses are correctly installed by using the following command and examining the output:

    `system license show`

    You can compare the output with the output that you captured in the section Prepare the nodes for upgrade.

12. If NetApp Storage Encryption (NSE) was in use on the configuration and you set the `setenv bootarg.storageencryption.support` command to `true` with the *`kmip.init.maxwait`* variable `off` (in Install and boot node4, Step 27), you need to reset the variable:

    `set diag; systemshell -node node_name -command sudo kenv -u -p kmip.init.maxwait`

13. Configure the SPs by using the following command on both nodes:

    `system service-processor network modify -node node_name`

    Refer to References to link to the *System Administration Reference* for information about the SPs and the *ONTAP 9.8 Commands: Manual Page Reference* for detailed information about the system `service-processor network modify` command.

14. Take the following actions on one of the new nodes:
    a. Enter advanced privilege level:

    ```
    set -privilege advanced
    ```

    b. Enter the following command:

    ```
    storage failover modify -node node_name -cifs- ndo-duration
    default|medium|low
    ```

    - Enter `medium` if the system will have workloads in which 50% to 75% of the operations will be 4 KB or smaller.
    - Enter `low` if the system will have workloads in which 75% to 100% of the operations will be 4 KB or smaller.

    c. Return to the admin level:

    ```
    set -privilege admin
    ```

    d. Reboot the system to ensure that the changes take effect.
15. If you want to set up a switchless cluster on the new nodes, refer to References to link to the *NetApp Support Site* and follow the instructions in *Transitioning to a two-node switchless cluster*.

**After you finish**

If Storage Encryption is enabled on node3 and node4, complete the section Set up Storage Encryption on the new controller module. Otherwise, complete the section Decommission the old system.

# Set up Storage Encryption on the new controller module

If the replaced controller or the HA partner of the new controller uses Storage Encryption, you must configure the new controller module for Storage Encryption, including installing SSL certificates and setting up key management servers.

**About this task**

This procedure includes steps that are performed on the new controller module. You must enter the command on the correct node.

**Steps**

1. Verify that the key management servers are still available, their status, and their authentication key information:

    ```
    security key-manager external show-status

    security key-manager onboard show-backup
    ```

2. Add the key management servers listed in the previous step to the key management server list in the new controller.
    a. Add the key management server:

    ```
    security key-manager external add-servers -key-servers
    key_management_server_ip_address
    ```

b. Repeat the previous step for each listed key management server. You can link up to four key management servers.

c. Verify the that the key management servers were added successfully:

```
security key-manager external show
```

3. On the new controller module, run the key management setup wizard to set up and install the key management servers.

   You must install the same key management servers that are installed on the existing controller module.

   a. Launch the key management server setup wizard on the new node:

   ```
   security key-manager external enable
   ```

   b. Complete the steps in the wizard to configure key management servers.

4. Restore authentication keys from all linked key management servers to the new node:

   ```
   security key-manager external restore -node new_controller_name
   ```

# Set up NetApp Volume Encryption on the new controller module

If the replaced controller or high availability (HA) partner of the new controller uses NetApp Volume Encryption (NVE), you must configure the new controller module for NVE.

**About this task**

This procedure includes steps that are performed on the new controller module. You must enter the command on the correct node.

**Steps**

1. Verify that the key management servers are still available, their status, and their authentication key information:

   ```
   security key-manager key query -node node
   ```

2. Add the key management servers listed in the previous step to the key management server list in the new controller:

   a. Add the key management server:

   ```
   security key-manager external add-servers -key-servers
   key_management_server_ip_address
   ```

   b. Repeat the previous step for each listed key management server. You can link up to four key management servers.

   c. Verify the that the key management servers were added successfully:

   ```
   security key-manager external show
   ```

3. On the new controller module, run the key management setup wizard to set up and install the key management servers.

   You must install the same key management servers that are installed on the existing controller module.

   a. Launch the key management server setup wizard on the new node:

      ```
      security key-manager external enable
      ```

   b. Complete the steps in the wizard to configure key management servers.

4. Restore authentication keys from all linked key management servers to the new node:

| For… | Use this command… |
|---|---|
| External key management (EKM) | `security key-manager external restore`<br><br>This command needs the OKM passphrase |
| Onboard Key Manager | `security key-manager onboard sync` |

**After you finish**

Check if any volumes were taken offline because authentication keys were not available or EKM servers could not be reached. Bring those volumes back online by using the `volume online` command.

# Decommission the old system

After upgrading, you can decommission the old system through the NetApp Support Site. Decommissioning the system tells NetApp that the system is no longer in operation and removes it from support databases.

**Steps**

1. Refer to References to link to the *NetApp Support Site* and log in.
2. Select **Products > My Products** from the menu.
3. On the **View Installed Systems** page, choose which **Selection Criteria** you want to use to display information about your system.

   You can choose one of the following to locate your system:

   - Serial Number (located on the back of the unit)
   - Serial Numbers for My Location

4. Select **Go!**

   A table displays cluster information, including the serial numbers.

5. Locate the cluster in the table and select **Decommission this system** from the Product Tool Set drop-down menu.

# Resume SnapMirror operations

You can resume SnapMirror transfers that were quiesced before upgrade and resume the SnapMirror relationships. The updates are on schedule after the upgrade is completed.

**Steps**

1. Verify the SnapMirror status on the destination:

   ```
   snapmirror show
   ```

2. Resume the SnapMirror relationship:

   ```
   snapmirror resume -destination-vserver vserver_name
   ```

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